THE SLICE OF A SMOOTH MORPHISM BY A FIBREWISE NONZERODIVISOR IS SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10774-10789 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10791-10843, the complete author proof (51 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smooth morphisms and the differential criterion, sheaves of differentials under quotients by a regular element, and nonzerodivisors on fibres; the computation is retained.

Source reference. Original lemma algebra-lemma-one-equation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma divisors-lemma-relative-Cartier: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-slice-smooth-once: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10661–10674. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-closed-point-fibre-locally-finite-type: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-dimension-fibre-at-a-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-locally-quasi-finite-rel-dimension-0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected

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spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-quasi-finite-points-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma schemes-lemma-fibre-topological: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10843–10843. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ be a point with image $s \in S$. Assume*

(1) *f is smooth at x, and*
(2) *x is a closed point of X_s and $\kappa(s) \subset \kappa(x)$ is separable.*

Then there exists an immersion $Z \to X$ containing x such that

(1) *$Z \to S$ is étale, and*
(2) *$Z_s = \{x\}$ set theoretically.*

Proof. We may and do replace S by an affine open neighbourhood of s. We may and do replace X by an affine open neighbourhood of x such that $X \to S$ is smooth. We will prove the lemma for smooth morphisms of affines by induction on $d = \dim_x(X_s)$.

The case $d = 0$. In this case we show that we may take Z to be an open neighbourhood of x. Namely, if $d = 0$, then $X \to S$ is quasi-finite at x, see Morphisms, Lemma morphisms-lemma-locally-quasi-finite-rel-dimension-0. Hence there exists an affine open neighbourhood $U \subset X$ such that $U \to S$ is quasi-finite, see Morphisms, Lemma morphisms-lemma-quasi-finite-points-open. Thus after replacing X by U we see that X is quasi-finite and smooth over S, hence smooth of relative dimension 0 over S, hence étale over S. Moreover, the fibre X_s is a finite discrete set. Hence after replacing X by a further affine open neighbourhood of X we see that $f^{-1}(\{s\}) = \{x\}$ (because the topology on X_s is induced from the topology on X, see Schemes, Lemma schemes-lemma-fibre-topological). This proves the lemma in this case.

Next, assume $d > 0$. Note that because x is a closed point of its fibre the extension $\kappa(x)/\kappa(s)$ is finite (by the Hilbert Nullstellensatz, see Morphisms, Lemma morphisms-lemma-closed-point-fibre-locally-finite-type). Thus we see $\Omega_{\kappa(x)/\kappa(s)} = 0$ as this holds for algebraic separable field extensions. Thus we may apply Lemma lemma-slice-smooth-once to find a diagram

$$\begin{array}{ccccc} D & \longrightarrow & U & \longrightarrow & X \\ & \searrow & & \swarrow & \downarrow \\ & & S & & \end{array}$$

with $x \in D$. Note that $\dim_x(D_s) = \dim_x(X_s) - 1$ for example because $\dim(\mathcal{O}_{D_s,x}) = \dim(\mathcal{O}_{X_s,x}) - 1$ by Algebra, Lemma algebra-lemma-one-equation (also $D_s \subset X_s$ is effective Cartier, see Divisors, Lemma divisors-lemma-relative-Cartier) and then using Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point. Thus the morphism $D \to S$ is smooth with $\dim_x(D_s) = \dim_x(X_s) - 1 = d - 1$. By induction hypothesis we can find an immersion $Z \to D$ as desired, which finishes the proof. □